STPM 2018 PHYSICS 960/2 (U2)

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $m_p = 1.67 \times 10^{-27}$ kg, $m_e = 9.11 \times 10^{-31}$ kg, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9\ N\,m^2\,C^{-2}$, $\mu_0 = 4\pi \times 10^{-7}\ H\,m^{-1}$.

Section A [15 marks]

1. A charge of +3.5 C is attracted to another charge, Q, at a distance of 0.72 cm. If the force of attraction is 4.86×10^{15} N, what is the value of Q?

A -8.0×10^3 C B -8.0 C C $+8.0$ C D $+8.0 \times 10^3$ C

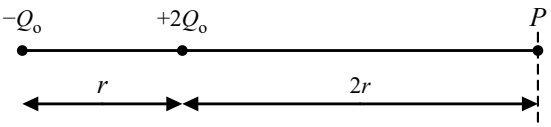
Answer: B

$$|Q| = \frac{Fr^2}{kq} = \frac{(4.86 \times 10^{15})(7.2 \times 10^{-3})^2}{(9.0 \times 10^9)(3.5)} = 8.0 \text{ C}$$

The force is attractive, so Q is negative: −8.0 C.

Explanation: Convert 0.72 cm to metres.

2. Two point charges, $-Q_o$ and $+2Q_o$, with a distance, r, and point P are shown in the diagram below.



What is the magnitude of the resultant electric field strength at point P, in terms of π, ε_0, Q_o and r?

A $\frac{Q_o}{6\pi\varepsilon_o r}$ B $\frac{Q_o}{3\pi\varepsilon_o r}$ C $\frac{7Q_o}{72\pi\varepsilon_o r^2}$ D $\frac{11Q_o}{72\pi\varepsilon_o r^2}$

Answer: C

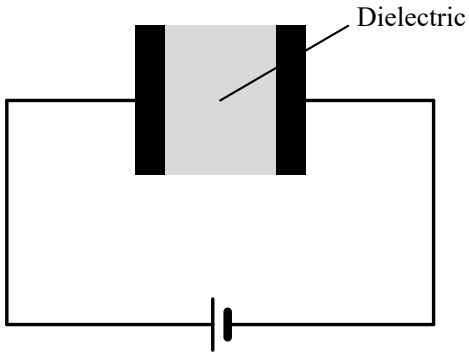
From $+2Q_o$ (distance $2r$, away): $\frac{2Q_o}{4\pi\varepsilon_o(2r)^2} = \frac{Q_o}{8\pi\varepsilon_o r^2}$

From $-Q_o$ (distance $3r$, towards it): $\frac{Q_o}{36\pi\varepsilon_o r^2}$

$$E = \frac{Q_o}{8\pi\varepsilon_o r^2} - \frac{Q_o}{36\pi\varepsilon_o r^2} = \frac{7Q_o}{72\pi\varepsilon_o r^2}$$

Explanation: The two fields at P point in opposite directions.

3. A capacitor of two parallel conducting plates which are separated by a piece of dielectric, is fully charged by a battery as shown in the diagram below.



The capacitance of the capacitor will change if

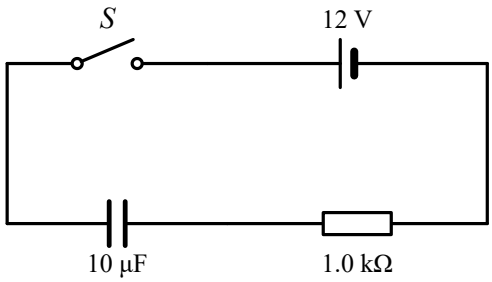
A different dielectric is used
B the capacitor is not fully charged
C the battery of different voltage is used
D the conducting plates of different metals are used

Answer: A

$C = \frac{\varepsilon_r \varepsilon_0 A}{d}$: a different dielectric (different ε_r) changes C

Explanation: The charge, voltage and plate metal do not affect the capacitance.

4. A battery of e.m.f. of 12 V is connected in series with a capacitor and a resistor as shown in the circuit below.



What is the charge in the capacitor 10 ms after S is closed?

A 330 μC B 210 μC C 76 μC D 44 μC

Answer: C

$$\tau = RC = (1.0 \times 10^3)(10 \times 10^{-6}) = 10 \text{ ms}; \; Q_0 = CV = 120 \text{ μC}$$

$$Q = Q_0(1 - e^{-1}) = 120(0.632) = 76 \text{ μC}$$

Explanation: After one time constant the capacitor is 63% charged.

5. Current flows in a metallic conductor when a potential difference is applied to the conductor. Which statement is true about the electrical conductivity in the conductor?

A The number of charge carriers increases.
B The average distance between the charge carriers increases.
C The charge carriers accelerate to the region of high potential.
D The charge carriers drift from the region of high potential to that of low potential.

Answer: C

Between collisions the free electrons (negative) accelerate towards the region of higher potential (C).

Explanation: D describes conventional current, not the electrons; the number of carriers does not change.

6. Which is true about current density?

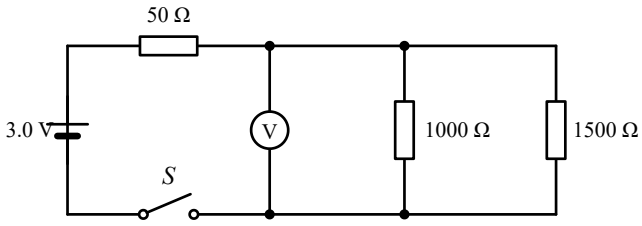
A Defined as current per unit volume
B Defined as current per unit length of a conductor
C Directly proportional to the electric field for ohmic material
D The ratio of the current density to the electric field is always constant at all temperature

Answer: C

$J = \sigma E$: for an ohmic material J is directly proportional to E (C)

Explanation: J is current per unit cross-sectional area, and σ changes with temperature.

7. A circuit is set up as shown in the diagram below.



When S is closed, the voltmeter reading is 2.76 V. What is the internal resistance of the dry cell?

A 2.0 Ω B 4.0 Ω C 37 Ω D 170 Ω

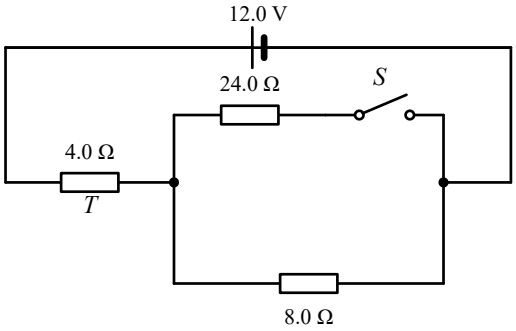
Answer: A

$$1000\ \Omega \parallel 1500\ \Omega = 600\ \Omega;\ I = \frac{2.76}{600} = 4.6 \times 10^{-3}\ \text{A}$$

$$\varepsilon = I(50 + 600 + r) \Longrightarrow r = \frac{3.0}{4.6 \times 10^{-3}} - 650 \approx 2\ \Omega$$

Explanation: The voltmeter reads the p.d. across the parallel pair.

8. A circuit contains a cell of e.m.f. of 12.0 V, a switch, *S*, and three resistors is shown in the diagram below.



What is the ratio of the dissipated power in resistor *T* when *S* is closed to that when *S* is opened?

A 7.8 B 2.3 C 1.4 D 1.2

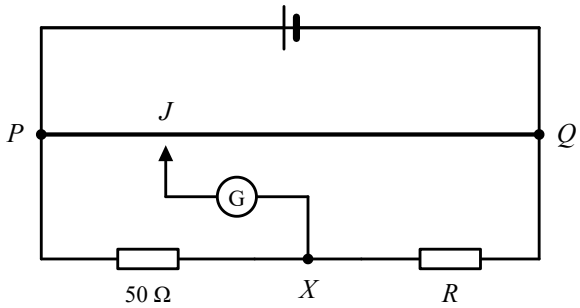
Answer: C

$$S \text{ open: } I = \frac{12.0}{4.0 + 8.0} = 1.0 \text{ A; } S \text{ closed: } 24 \parallel 8.0 = 6.0\ \Omega,\ I = \frac{12.0}{10.0} = 1.2 \text{ A}$$

$$\text{Ratio} = \frac{(1.2)^2(4.0)}{(1.0)^2(4.0)} = 1.44 \approx 1.4$$

Explanation: Closing *S* lowers the total resistance, so more current flows through *T*.

9. A potentiometer is used to determine the resistance of a resistor, *R*, is shown in the diagram below.



The length of the uniform slide wire, *PQ*, is 100 cm. If the balance length, *PJ*, is 25 cm, what is the value of *R*?

A 25 Ω B 50 Ω C 75 Ω D 150 Ω

Answer: D

$$\frac{50}{R} = \frac{PJ}{JQ} = \frac{25}{75} \Longrightarrow R = 150\ \Omega$$

Explanation: At balance the bridge ratios are equal.

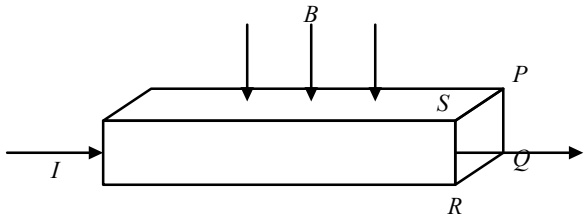
10. Which is **not** correct about the Hall effects?

A To measure the magnetic flux density
B To determine the drift velocity of the charges
C To determine the sign of charge carriers in a semiconductor
D To measure the properties of light in an electromagnet spectrum

Answer: D
The Hall effect measures B, the drift velocity and the sign of the carriers; it has nothing to do with the properties of light (D).

Explanation: It is a magnetic-field effect on moving charges.

11. A copper conductor carrying a current, I, is shown in the diagram below.



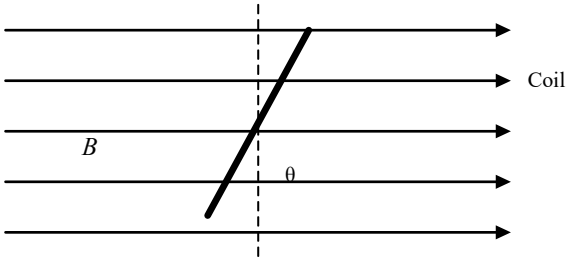
When a magnetic field, B, is applied vertically downward on the conductor, an electric field is produced from

A R to Q B P to Q C Q to R D Q to P

Answer: A
In copper the carriers are electrons, moving opposite to I. With **B** downwards the magnetic force pushes them towards the face containing Q, leaving the opposite face (R) positive.
The Hall field points from the positive face to the negative face: from R to Q.

Explanation: Apply Fleming's left-hand rule to the current, then reverse it for electrons.

12. A circular coil of radius r, which has number of turns, N, is placed in a uniform magnetic field, B. The coil is initially placed in the field so that the plane of the coil is perpendicular to the field. The coil is then rotated at an angle θ to the magnetic field as shown in the diagram below.



What is the change in the magnetic flux through the coil?

A $\pi r^2 NB \sin\theta$
B $\pi r^2 NB \cos\theta$
C $\pi r^2 NB (\sin\theta - 1)$
D $\pi r^2 NB (\cos\theta - 1)$

Answer: D

Initially $\Phi_1 = NB\pi r^2$ (plane $\perp$ **B**); finally $\Phi_2 = NB\pi r^2 \sin\theta$ (plane at θ to **B**)

$$\Delta\Phi = \pi r^2 NB(\sin\theta - 1)$$

Explanation: Flux depends on the component of **B** normal to the plane.

13. A current of 2.0 A flows through a pure inductor with an inductance of 200 mH. What is the energy stored in the inductor?

A 0.040 J B 0.20 J C 0.40 J D 0.80 J

Answer: C

$$U = \frac{1}{2}LI^2 = \frac{1}{2}(0.200)(2.0)^2 = 0.40 \text{ J}$$

Explanation: 200 mH = 0.200 H.

14. The root mean square value of an alternating current is

A half of the maximum value of the current
B an instantaneous maximum value of the current
C the value of a steady current that will charge a capacitor at the same rate
D the value of a steady current that produces the same heating effect in a resistor

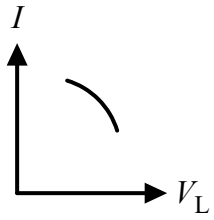
Answer: D

The r.m.s. current is the steady current that produces the same heating effect in a resistor (D).

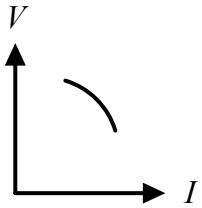
Explanation: For a sinusoidal current $I_{rms} = I_0/\sqrt{2}$.

15. An alternating voltage supply is connected to a pure inductor. Which phasor diagram is correct for the inductor voltage, V_L, and the current, I?

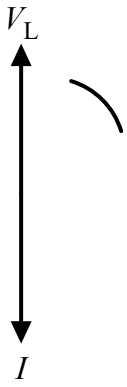
A



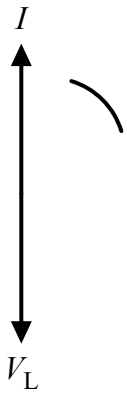
B



C



D



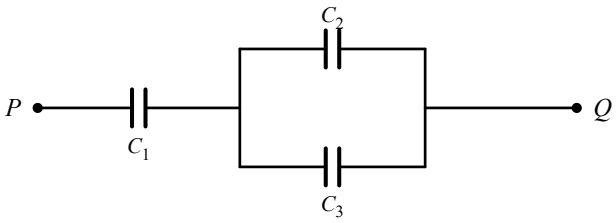
Answer: B

In a pure inductor the voltage leads the current by 90°: with anticlockwise rotation V is 90° ahead of I (B).

Explanation: "ELI": in an inductor (L), e.m.f. (E) leads current (I).

Section B [15 marks]

16. Three capacitors, C_1, C_2 and C_3, each labelled as 300 μF, 6.0 V are arranged as shown in the diagram below.



(a) Determine the total capacitance between P and Q. [3]

Answer

$C_2 \parallel C_3 = 300 + 300 = 600$ μF	1
In series with C_1: $\frac{(300)(600)}{300 + 600}$	1
$C = 200$ μF	1
	3

Explanation: Parallel first, then series.

(b) Determine the maximum potential difference between P and Q that can be safely applied. [4]

Answer

In series, C_1 and the parallel pair carry the same charge; C_1 (smaller capacitance) has the larger p.d. and reaches 6.0 V first — 1

$Q = (300 \times 10^{-6})(6.0) = 1.8 \times 10^{-3}$ C — 1

P.d. across the pair $= \dfrac{1.8 \times 10^{-3}}{600 \times 10^{-6}} = 3.0$ V — 1

Maximum $V_{PQ} = 6.0 + 3.0 = 9.0$ V — 1

4

Total for question **7**

Explanation: Not 18 V: the p.d. does not divide equally.

17. A lightning discharges a current density of 30 000 A m^{-2} for a duration of 10.0 ms. The potential difference between the bottom edge of a cloud and the surface of the ground is 1.00×10^8 V.

(a) Calculate the number of electrons flow per unit area for each lightning. [4]

Answer

Charge per unit area $= Jt$ — 1

$= (30\,000)(10.0 \times 10^{-3}) = 300$ C m^{-2} — 1

$n = \dfrac{Jt}{e} = \dfrac{300}{1.60 \times 10^{-19}}$ — 1

$n = 1.88 \times 10^{21}$ m^{-2} — 1

4

Explanation: Current density × time = charge per unit area.

(b) Air is an excellent insulator but it becomes an electric conductor when the electric fields strength exceeds 3.00×10^6 N C^{-1}. Determine the electrical conductivity of air between the cloud and the surface of the ground during lightning. [2]

Answer

$\sigma = \dfrac{J}{E} = \dfrac{30\,000}{3.00 \times 10^6}$ — 1

$\sigma = 1.0 \times 10^{-2}\ \Omega^{-1}$ m^{-1} — 1

2

Explanation: Use the breakdown field as the field in the air.

(c) Determine the energy transferred per unit area for each lightning. [2]

Answer

Energy per unit area = (charge per unit area) × V = (300)(1.00 × 10^8) 1

= 3.0 × 10^{10} J m^{-2} 1

2

Total for question **8**

Explanation: $W = QV$.

Section C [30 marks]

18. (a) Define Gauss's law. [1]

Answer

The total electric flux through a closed surface equals the net charge enclosed divided by ε_0:

$\Phi = \frac{Q}{\varepsilon_0}$ 1

1

Explanation: The closed surface is called a Gaussian surface.

(b) A hollow isolated conducting sphere with a radius of R carries positive charges with a surface charged density, σ.

(i) By using Gauss's law, derive an expression for the electric field strength, E, in terms of σ, r, R and ε_0 at a point of distance, r, from the centre of the sphere for $r < R$ and $r > R$. [5]

Answer

$r < R$: a Gaussian sphere inside encloses no charge 1

so $E(4\pi r^2) = 0 \Longrightarrow E = 0$ 1

$r > R$: charge enclosed $Q = \sigma(4\pi R^2)$ 1

$E(4\pi r^2) = \frac{\sigma(4\pi R^2)}{\varepsilon_0}$ 1

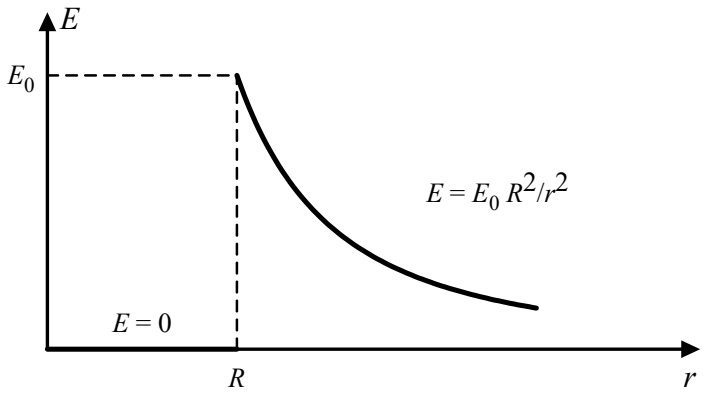
$E = \frac{\sigma R^2}{\varepsilon_0 r^2}$ 1

5

Explanation: By symmetry E is radial and has the same magnitude everywhere on the Gaussian sphere.

(ii) Sketch a graph of E against r. [4]

Answer

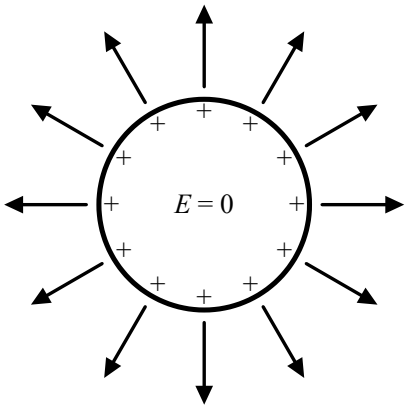


$E = 0$ for $r < R$	1
Maximum value at $r = R$	1
$E_0 = \sigma/\varepsilon_0$ marked at $r = R$	1
Decreasing as $1/r^2$ for $r > R$	1
	4

Explanation: At $r = R$, $\sigma R^2/\varepsilon_0 R^2 = \sigma/\varepsilon_0$.

(iii) Draw a diagram to show the charge distribution and the electric field lines produced by the sphere. [2]

Answer



Positive charges distributed uniformly on the outer surface	1
Field lines radially outwards, perpendicular to the surface	1
	2

Explanation: No field lines inside the conductor.

(c) A Van de Graff generator consists of a metal sphere with a radius of 20.0 cm. The breakdown electric field in air is 3.00×10^6 N C^{-1}.

(i) Calculate the maximum quantity of charge which can be stored in the sphere. [2]

Answer

$Q = 4\pi\varepsilon_0 r^2 E = \frac{Er^2}{k} = \frac{(3.00 \times 10^6)(0.200)^2}{9.0 \times 10^9}$ 1

$Q = 1.33 \times 10^{-5}$ C 1

2

Explanation: The field is largest at the surface, so breakdown starts there.

(ii) State **one** way to increase the maximum quantity of charge to be stored in the Van de Graff generator. [1]

Answer

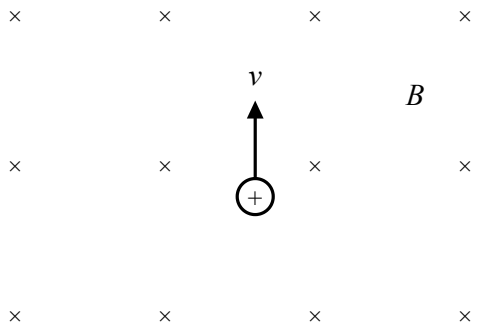
Use a sphere of larger radius (or surround it with a gas of higher breakdown field) 1

1

Total for question **15**

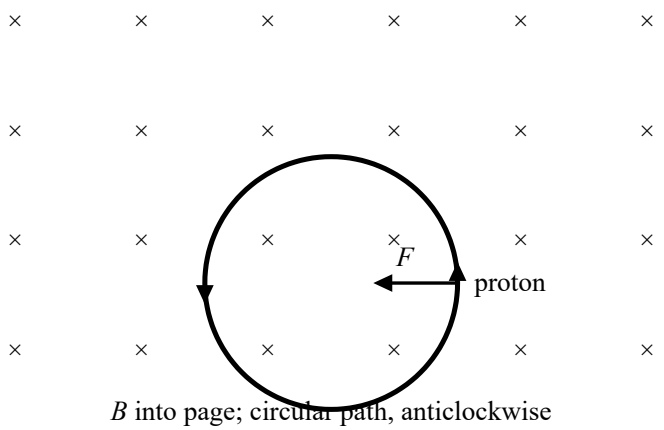
Explanation: $Q_{max} \propto r^2$.

19. (a) A proton moves in a direction perpendicular to a uniform magnetic field, B, which directs into the plane of this page as shown in the diagram below.



(i) Sketch the path of the proton and the direction of the magnetic force, F, which acts on the proton by the uniform magnetic field. [2]

Answer



B into page; circular path, anticlockwise

Circular path (anticlockwise)	1
Force F towards the centre: to the left at the starting point	1
	2

Explanation: Fleming's left-hand rule: **v** up, **B** into the page gives **F** to the left.

(ii) If the velocity of the proton is 1.0×10^5 m s^{-1} and the magnitude of the magnetic field is 0.75 T, calculate the radius of the path. [3]

Answer

$evB = \frac{mv^2}{r} \Longrightarrow r = \frac{mv}{eB}$ 1

$r = \frac{(1.67 \times 10^{-27})(1.0 \times 10^5)}{(1.60 \times 10^{-19})(0.75)}$ 1

$r = 1.39 \times 10^{-3}$ m 1

3

Explanation: The magnetic force provides the centripetal force.

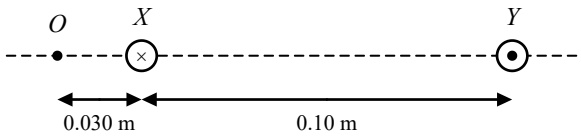
(iii) The proton is replaced by an electron, which moves with the same velocity in the magnetic field. Explain the differences in the path of the electron as compared to that of the proton. [4]

Answer

The electron has opposite charge, so the force is in the opposite direction (to the right)	1
It moves in a circle in the opposite sense (clockwise)	1
The electron's mass is much smaller ($m_e \ll m_p$) while $\|q\|$, v and B are the same	1
so its radius is much smaller: $r = m_e v/(eB) = 7.6 \times 10^{-7}$ m (about 1/1840 of the proton's)	1
	4

Explanation: It also completes each revolution in a much shorter time.

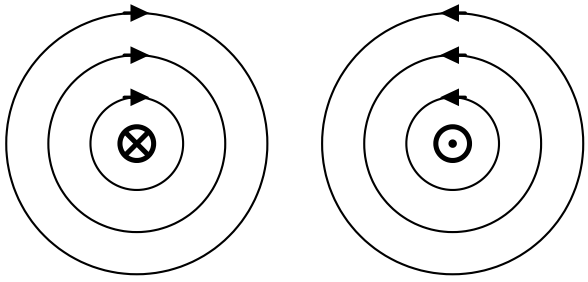
(b) Two long parallel wires, X and Y, are separated by a distance of 0.10 m as shown in the diagram below.



X carries current of 15.0 A into the page, while Y carries current of 18.0 A in the opposite direction. O is located at 0.030 m from X.

(i) Sketch the magnetic field pattern produced by the current in X and Y. [2]

Answer



X (into page): clockwise Y (out of page): anticlockwise

Concentric circles around each wire	1
Clockwise around X (into page) and anticlockwise around Y (out of page)	1
	2

Explanation: Right-hand grip rule; the circles are closer together near each wire.

(ii) Determine the magnitude and the direction of the resultant magnetic field at O. [4]

Answer

$B_X = \frac{\mu_0 I}{2\pi r} = \frac{(4\pi \times 10^{-7})(15.0)}{2\pi(0.030)} = 1.0 \times 10^{-4}$ T, up the page — 1

$B_Y = \frac{(4\pi \times 10^{-7})(18.0)}{2\pi(0.130)} = 2.77 \times 10^{-5}$ T, down the page — 1

Resultant = $1.0 \times 10^{-4} - 2.77 \times 10^{-5}$ — 1

$B = 7.2 \times 10^{-5}$ T, up the page — 1

4

Total for question **15**

Explanation: O is 0.030 + 0.10 = 0.130 m from Y.

20. (a) State Faraday's law and Lenz's law for electromagnetic induction. [2]

Answer

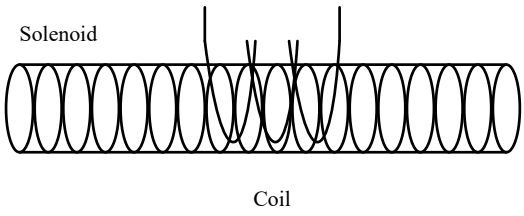
Faraday: the induced e.m.f. is directly proportional to the rate of change of magnetic flux linkage — 1

Lenz: the induced current flows in a direction that opposes the change producing it — 1

2

Explanation: $\varepsilon = -\mathrm{d}(N\Phi)/\mathrm{d}t$.

(b) A long solenoid is placed coaxially inside a short circular coil as shown in the diagram below.



The solenoid consists of 30 turns per cm with a cross-sectional area of 1.5 cm^2 while the coil has 50 turns. A current of 2.0 A flows clockwise in the solenoid.

(i) Determine the magnitude and direction of the magnetic field produced in the solenoid. [3]

Answer

n = 30 turns per cm = 3000 m^{-1} — 1

$B = \mu_0 nI = (4\pi \times 10^{-7})(3000)(2.0) = 7.54 \times 10^{-3}$ T — 1

Along the axis of the solenoid, directed away from the viewer who sees the current flowing clockwise (right-hand grip rule) — 1

3

Explanation: The field inside a long solenoid is uniform and axial.

(ii) Calculate the magnetic flux in the solenoid. [2]

Answer

$\Phi = BA = (7.54 \times 10^{-3})(1.5 \times 10^{-4})$ 1

$\Phi = 1.13 \times 10^{-6}$ Wb 1

2

Explanation: 1.5 cm^2 = 1.5×10^{-4} m^2; the same flux passes through the coil.

(iii) If the current in the solenoid decreases to zero in 0.10 s, determine the average e.m.f. induced in the coil. [2]

Answer

$\varepsilon = N\frac{\Delta\Phi}{\Delta t} = (50)\frac{1.13 \times 10^{-6}}{0.10}$ 1

$\varepsilon = 5.65 \times 10^{-4}$ V 1

2

Explanation: Use the 50 turns of the coil.

(iv) If the resistance of the coil is 0.50 Ω, determine the magnitude and direction of the current in the coil. [3]

Answer

$I = \frac{\varepsilon}{R} = \frac{5.65 \times 10^{-4}}{0.50}$ 1

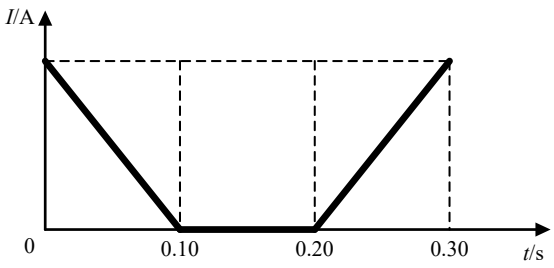
$I = 1.13 \times 10^{-3}$ A 1

Clockwise (same sense as the solenoid current), to oppose the decrease in flux 1

3

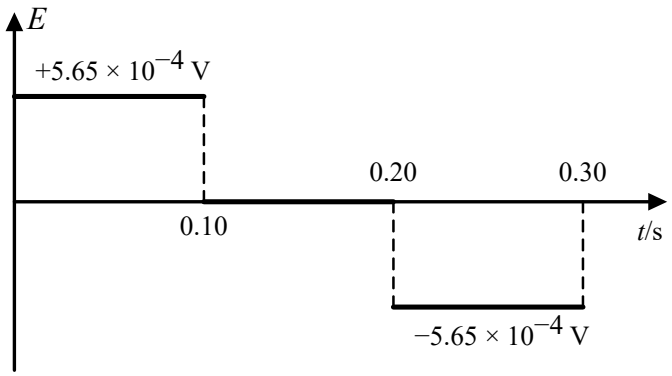
Explanation: Lenz's law: the induced current tries to maintain the original flux.

(c) The variation of current in the solenoid, I, with time, t, in (b) is shown in the graph below.



Sketch a graph to show the variation of induced e.m.f., E, in the coil with time, t. [3]

Answer



Constant (non-zero) e.m.f. from 0 to 0.10 s	1
Zero e.m.f. from 0.10 s to 0.20 s	1
Constant e.m.f. of equal magnitude but opposite sign from 0.20 s to 0.30 s	1
	3
Total for question	**15**

Explanation: $E \propto -\mathrm{d}I/\mathrm{d}t$: the gradients of the current graph are equal and opposite.